LAB 1 – Requirements Engineering & UML Use-Case Modelling

Problem Statement #65

Urban Rooftop Gardening Co-op App

A community urban agriculture app for registering rooftop plots, logging crop yields and harvest schedules, sharing soil maintenance tips, and trading surplus produce.

Course Deliverable	Requirements Table + UML Use-Case Diagram + Use-Case Flow
Actors	Co-op Gardener, Community Member, Community Leader
Functional Requirements	FR-001 to FR-005
Non-Functional Requirements	NFR-001 to NFR-002

Prepared for Lab 1 submission

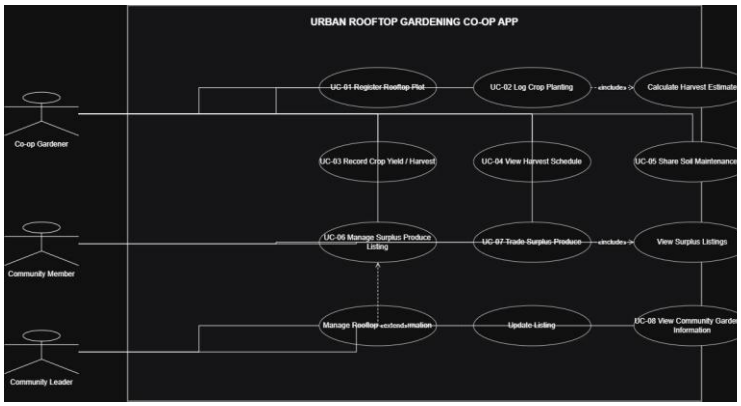
1. Complete Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow gardeners to log seasonal crop planting dates, computing estimated harvest windows and expected yield weight.	High	Pass: A valid planting date generates an estimated harvest window and expected yield. Fail: An invalid planting date sequence is accepted.	Helps gardeners plan harvesting and estimate production.
FR-002	Functional	The system shall allow co-op gardeners to register and manage rooftop plots assigned to them.	High	Pass: A gardener can enter plot details and successfully view the registered plot. Fail: Incomplete or invalid plot information is accepted.	Maintains an organized record of rooftop plots.
FR-003	Functional	The system shall allow gardeners to record crop yields and completed harvest quantities for their registered plots.	High	Pass: A crop and harvest quantity can be saved and appears in plot history. Fail: A negative or invalid quantity is accepted.	Provides accurate information about crop production.
FR-004	Functional	The system shall allow co-op members to share and view soil maintenance tips.	Medium	Pass: A submitted tip becomes visible in the soil-maintenance section. Fail: An empty tip can be submitted.	Encourages knowledge sharing among community gardeners.
FR-005	Functional	The system shall allow gardeners to create, view, and update surplus produce listings for trading with other community members.	High	Pass: A valid listing appears on the trade board and can be updated or removed by its owner. Fail: Invalid or incomplete listings are published.	Supports surplus produce trading.
NFR-001	Nonfunctional – Performance & Security	The produce trade board must update available surplus listings in real-time across community members.	High	Pass: Listing changes are reflected for other users within the defined target latency during simulated peak load. Fail: Updates exceed the target latency or security testing fails.	Ensures members see current produce availability.
NFR-002	Nonfunctional – Usability	The system shall allow a registered gardener to complete a common task, such as adding a harvest record, within 3 minutes without assistance.	Medium	Pass: At least 90% of test users complete the task within 3 minutes. Fail: Fewer than 90% complete it within the limit.	Makes the app practical for users with different technical experience.

View Community Gardening Info

2. UML Use-Case Diagram

The diagram contains the three actors, primary use cases, and the required «include» and «extend» relationships.



3. Use-Case Flow Specification

Use Case: UC-03 – Record Crop Yield / Harvest

Primary Actor: Co-op Gardener

Description: Allows a gardener to record the quantity of crops harvested from a registered rooftop plot.

Preconditions

• The gardener is registered with the co-op.

• The gardener has access to the application.

• At least one rooftop plot has been registered.

• A crop has been associated with the selected plot.

Postconditions

• The harvest record is stored in the system.

• The crop yield history is updated.

• The recorded harvest quantity is associated with the selected plot and crop.

Main Success Scenario

1. Co-op Gardener opens the Record Crop Yield / Harvest option.

2. System displays the gardener's registered rooftop plots.

3. Gardener selects the relevant rooftop plot.

4. System displays the crops associated with the selected plot.

5. Gardener selects the harvested crop.

6. System displays a form requesting the harvest date and harvested quantity.

7. Gardener enters valid harvest information.

8. System validates the entered information.

9. System stores the harvest record.

10. System updates the crop yield history.

11. System displays a confirmation that the harvest has been successfully recorded.

12. Use case ends successfully.

Alternate Flow – 8A: Invalid Harvest Quantity

8A1. System detects that the entered harvest quantity is invalid, such as a negative value.

8A2. System displays an error message asking the gardener to enter a valid quantity.

8A3. Gardener enters a corrected quantity.

8A4. System validates the corrected information.

8A5. System stores the harvest record and continues from Step 9 of the Main Success Scenario.